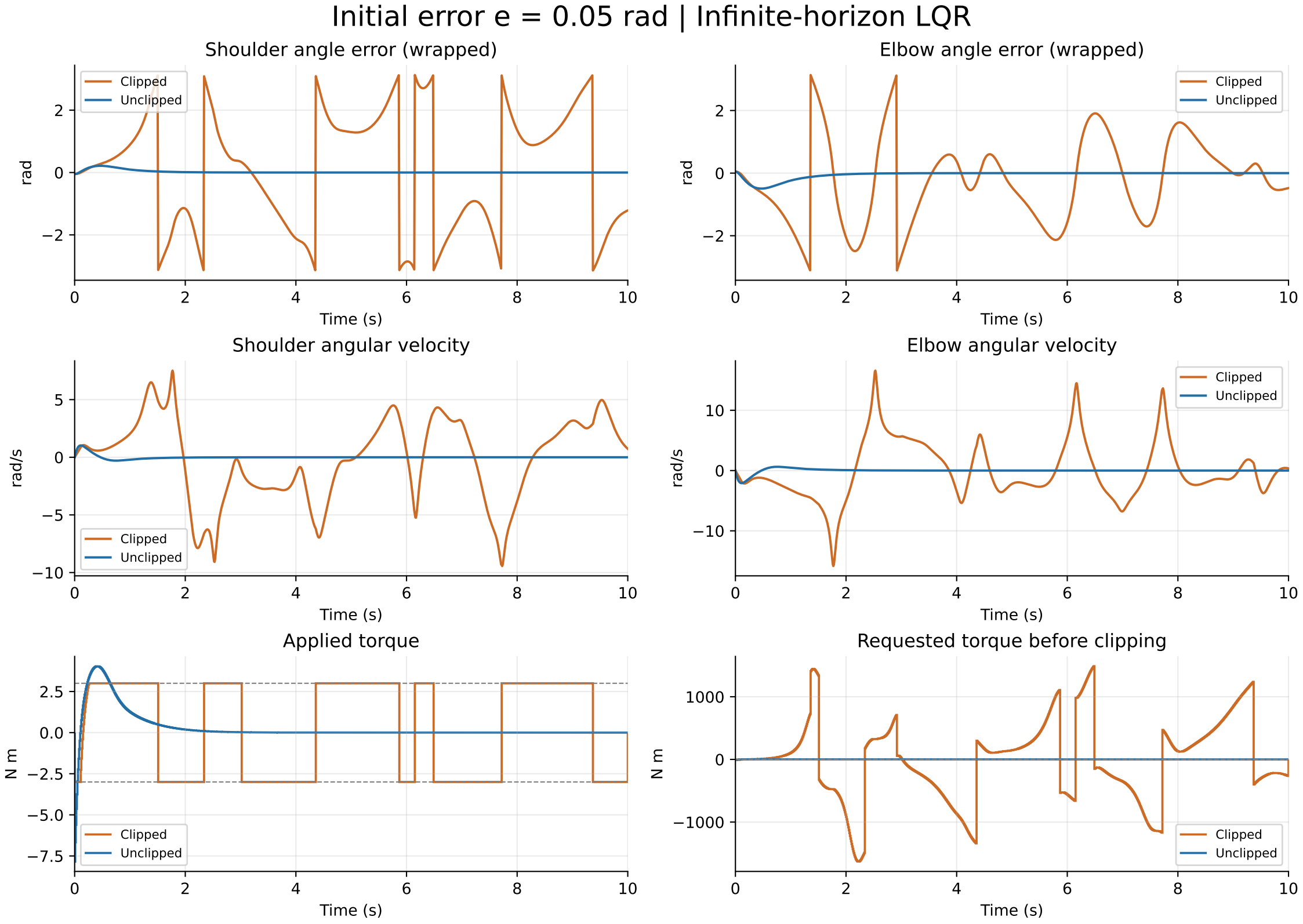
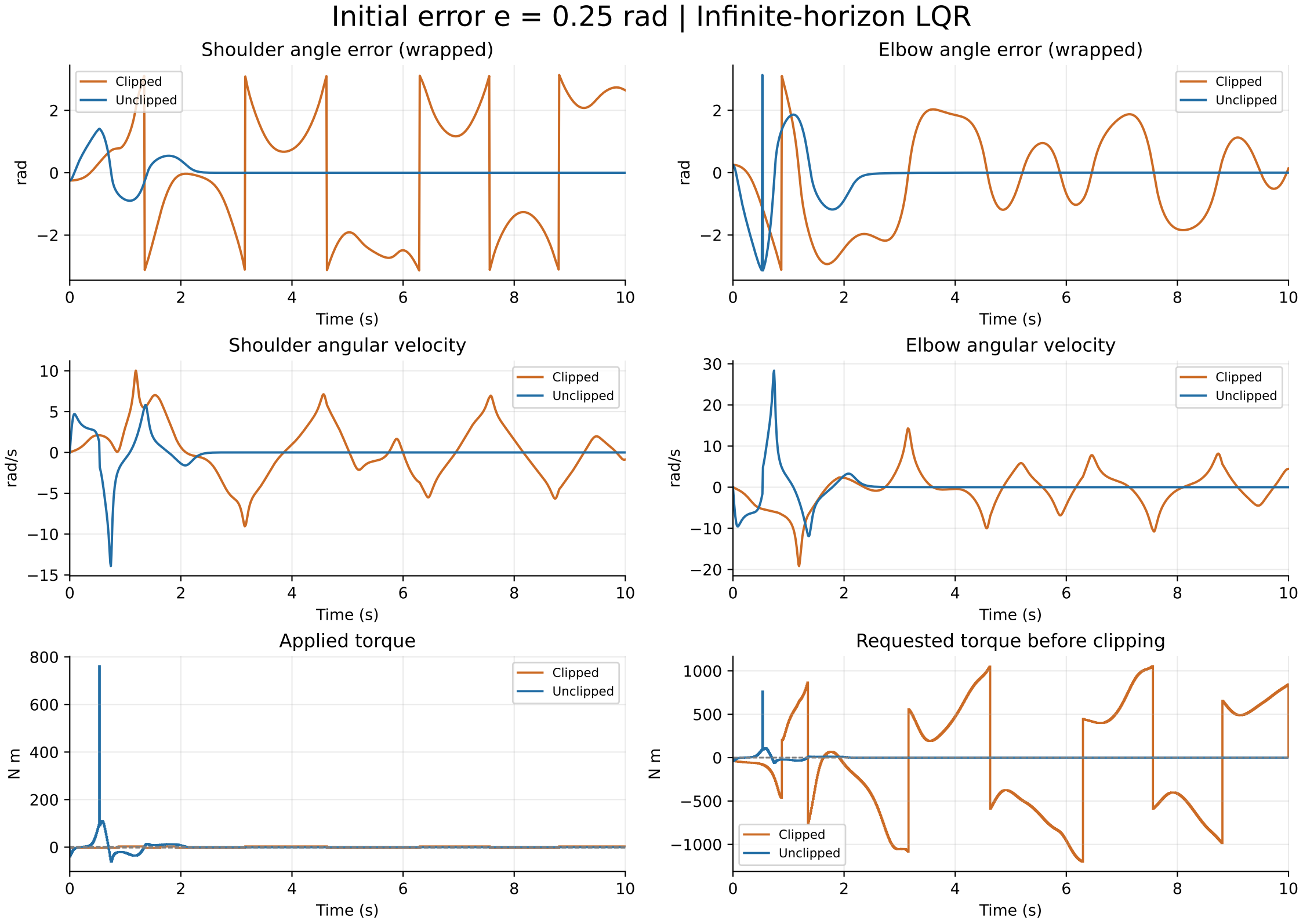
Acrobot | Infinite-horizon LQR

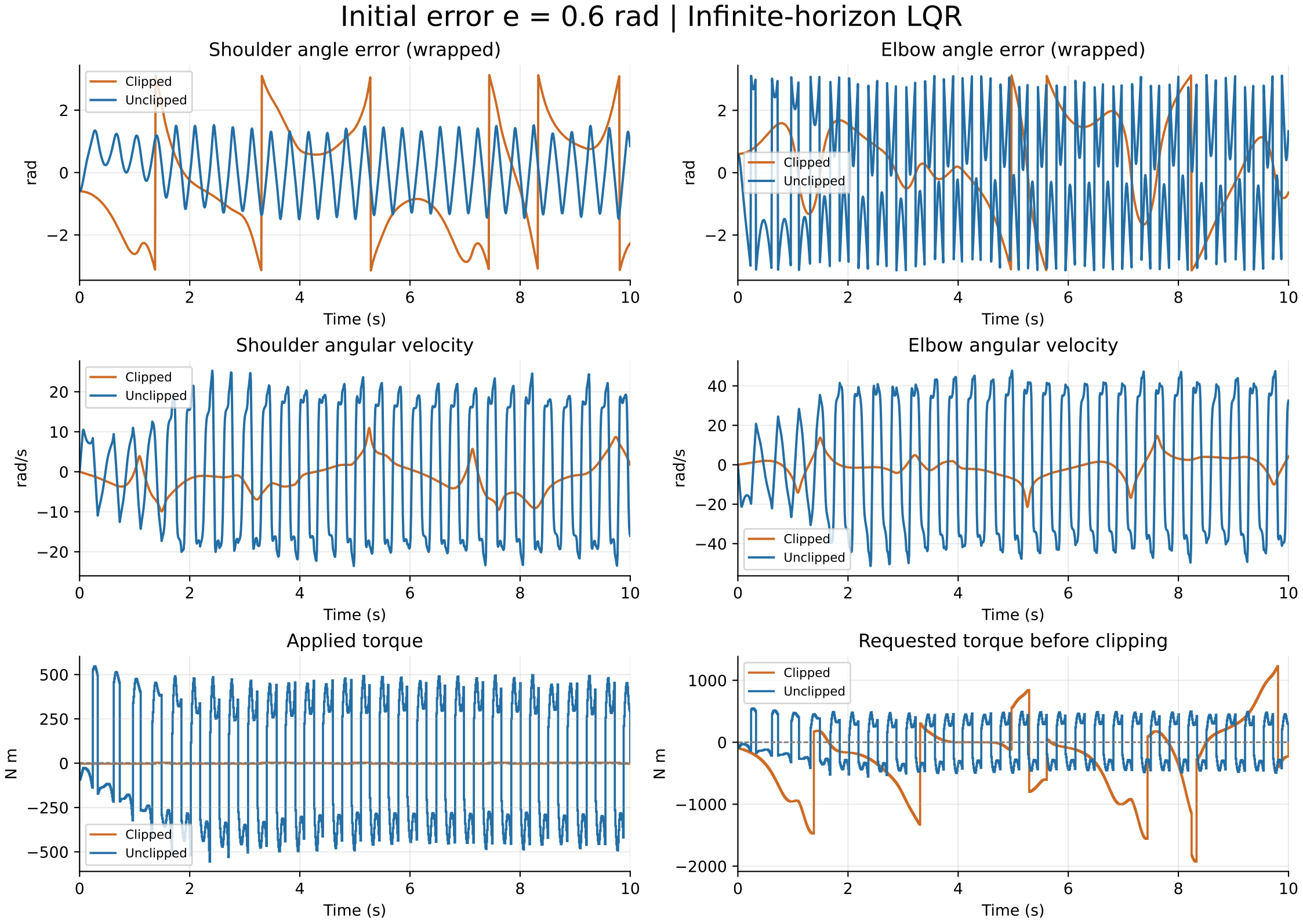
Six nonlinear Drake simulations | 10 s | control period 0.01 s
Initial state: [pi - e, e, 0, 0], e = 0.05, 0.25, 0.6 rad.
Clipped torque: +/-3 N m. Unclipped: no torque limit.
Q = diag(10, 10, 1, 1); R = 1; u = -K error (angles wrapped).
One constant gain is used for every run; saturation does not redesign K.
K = [-262.131925 -105.190703 -112.642345 -53.383978]

Local closed-loop eigenvalues (discrete time):
Euler design A - BK: 0.902955+0.000000j, 0.958624+0.000000j, 0.977823+0.000205j, 0.977823-0.000205j
Exact zero-order hold: 0.900450+0.000000j, 0.957031+0.000000j, 0.979234+0.000000j, 0.977844+0.000000j
Spectral radii: design 0.977823; exact 0.979234.
These poles apply near upright, where clipping is inactive.
Saturated nonlinear motion has no single fixed closed-loop pole set.

e (rad)	Clipping	Peak |u| (N m)	Clipped steps	Final angle norm (rad)	Final speed norm (rad/s)
0.05	On	3.000	985	1.31	0.82
0.05	Off	7.847	0	2.34e-09	4.87e-09
0.25	On	3.000	998	2.65	4.53
0.25	Off	760.988	0	5.37e-09	1.13e-08
0.6	On	3.000	931	2.35	4.58
0.6	Off	555.376	0	1.58	36.2


Initial error e = 0.05 rad | Infinite-horizon LQR
Shoulder angle error (wrapped)
Elbow angle error (wrapped)
Shoulder angular velocity
Elbow angular velocity
Applied torque
Requested torque before clipping
Clipped
Unclipped
rad
rad/s
N m
Time (s)


Initial error e = 0.25 rad | Infinite-horizon LQR
Shoulder angle error (wrapped)
Elbow angle error (wrapped)
Shoulder angular velocity
Elbow angular velocity
Applied torque
Requested torque before clipping
Clipped
Unclipped
rad
rad/s
N m
Time (s)


Initial error e = 0.6 rad | Infinite-horizon LQR
Shoulder angle error (wrapped)
Elbow angle error (wrapped)
Shoulder angular velocity
Elbow angular velocity
Applied torque
Requested torque before clipping
Clipped
Unclipped
rad
rad/s
N m
Time (s)